Mathematical Foundations for Machine Learning

AIML ZC416 — Consolidated Lecture Notes: Vector Spaces, Analytic Geometry & Matrix Decomposition

Compiled from Prof. Saurabh's live class recordings, BITS Pilani WILP

Sessions 2 – 4 (with a Session 5 preview) September 8, 2026

Contents

About These Notes

These notes are compiled directly from **two live class recordings** of *Mathematical Foundations for Machine Learning (AIML ZC416)* taught by Prof. Saurabh, and are cross-mapped to the official **Session Plan** in your course handout. Every concept below is explained the way it was taught in class, then reinforced with a **worked example**, a **solved practice question**, and a note on **where it shows up in real ML systems**.

The two recordings, together, cover:

Covered live in class	Handout Session	Handout Module
Linear combinations, span, linear independence, basis, dimension, rank	**Session 2**	M2
Norms, dot product, inner product, positive definiteness, Cauchy–Schwarz, orthogonality, orthonormal matrices, Gram–Schmidt (intro)	**Session 3**	M2
Determinants, eigenvalues & eigenvectors, the Power Method	**Session 4**	M3
(Preview only, taught in detail next session) Diagonalization, Spectral Decomposition, SVD, low-rank approximation	**Session 5**	M3

A full, fully-solved **Practice Problem Set for Lectures 4 and 5** (matrix decompositions) is reproduced in **Appendix A** at the end of this document, exactly as provided for your practice.

PART I — Session 2: Vector Spaces

Syllabus Link

Handout Session 2: *"Vector Spaces – linear independence, basis and rank, affine spaces"* (Module M2, Textbook T1 Sec 2.4–2.6, 2.8).

Everything in linear algebra, Prof. Saurabh stressed at the very start, comes down to just **two operations**: *scalar multiplication* and *vector addition*. Every idea in this section is built out of those two operations alone.

1.1 Linear Combinations

Concept

Given vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ and scalars $\lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{R}$, a **linear combination** is

$$\lambda_1\mathbf{v}_1 + \lambda_2\mathbf{v}_2 + \cdots + \lambda_n\mathbf{v}_n.$$

Geometrically: scalar multiplication stretches/shrinks/flips a vector *along its own line through the origin*; vector addition places vectors tip-to-tail. A linear combination does both at once.

Worked Example

Take $\mathbf{v} = (2, 3)$. Scaling by λ gives $(2\lambda, 3\lambda)$ — this always lands back on the same line through the origin, no matter what real value λ takes (this is exactly why one vector alone can never escape its own line).
Now take two vectors, $\mathbf{v}_1 = (1, 0)$ and $\mathbf{v}_2 = (0, 1)$, with $\lambda_1 = 3,\ \lambda_2 = -2$:

$$3(1, 0) + (-2)(0, 1) = (3, -2).$$

By choosing different λ_1, λ_2 you can reach *any* point in $\mathbb{R}^2$ — this is the seed of the idea of **span**, covered next.

Solved Question

Q. Given $\mathbf{v}_1 = (1, 2)$ and $\mathbf{v}_2 = (3, 1)$, find scalars α, β such that $\alpha\mathbf{v}_1 + \beta\mathbf{v}_2 = (7, 5)$.
Solution. This gives two equations: $\alpha + 3\beta = 7$ and $2\alpha + \beta = 5$. From the second, $\beta = 5 - 2\alpha$. Substituting: $\alpha + 3(5 - 2\alpha) = 7 \Rightarrow \alpha + 15 - 6\alpha = 7 \Rightarrow -5\alpha = -8 \Rightarrow \alpha = 1.6,\ \beta = 1.8$.
Check: $1.6(1, 2) + 1.8(3, 1) = (1.6, 3.2) + (5.4, 1.8) = (7, 5)$. ✓
Answer: $\alpha = 1.6,\ \beta = 1.8$.

Real-World / ML Application

Every forward pass of a neural network layer, $\mathbf{y} = W\mathbf{x}$, is literally a linear combination of the input features, weighted by the learned weights. Portfolio return in finance is a linear combination of individual asset returns weighted by portfolio allocation. RGB colour mixing (as shown in class using the CNN Explainer visualization) is a linear combination of the red, green and blue channel intensities.

1.2 Span

Concept

The **span** of a set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is the set of *all* possible linear combinations of them — i.e., every new point you can reach by scaling and adding these vectors.

- Two *independent* vectors in $\mathbb{R}^2$ span all of $\mathbb{R}^2$.
- Two *collinear (dependent)* vectors span only the single line they both lie on.

Worked Example

$\mathbf{v}_1 = (1, 2)$ and $\mathbf{v}_2 = (2, 4)$: since $\mathbf{v}_2 = 2\mathbf{v}_1$, they are dependent, and their span is only the line $y = 2x$ — you can never leave that line by combining them, no matter what scalars you pick.
Contrast with $\mathbf{v}_1 = (1, 0),\ \mathbf{v}_2 = (0, 1)$: independent, and their span is the entire plane $\mathbb{R}^2$.

Solved Question

Q. Do $\mathbf{v}_1 = (1, 1, 0),\ \mathbf{v}_2 = (0, 1, 1),\ \mathbf{v}_3 = (1, 2, 1)$ span $\mathbb{R}^3$?
Solution. Check whether $\mathbf{v}_3$ is already reachable from $\mathbf{v}_1, \mathbf{v}_2$:

$$\mathbf{v}_1 + \mathbf{v}_2 = (1, 1, 0) + (0, 1, 1) = (1, 2, 1) = \mathbf{v}_3.$$

So $\mathbf{v}_3$ adds *no new direction* — it is already inside the span of $\mathbf{v}_1, \mathbf{v}_2$.
Answer: No. The three vectors only span the 2-D plane spanned by $\mathbf{v}_1, \mathbf{v}_2$ (a flat plane through the origin sitting inside $\mathbb{R}^3$), not the full three-dimensional space.

Real-World / ML Application

The professor's own Netflix/e-commerce analogy: if, out of a million users, only 100 truly independent "taste directions" exist, then the span of just those 100 vectors already contains every one of the million user-preference vectors. This is precisely the intuition behind matrix-factorization-based recommender systems (and previews the SVD/low-rank ideas in Session 5). Similarly, word-embedding arithmetic ("king − man + woman ≈ queen") works because meaningful semantic directions combine linearly to reach new, meaningful points in embedding space.

1.3 Linear Independence and Dependence

Concept

Vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ are **linearly independent** if

$$\lambda_1\mathbf{v}_1 + \lambda_2\mathbf{v}_2 + \cdots + \lambda_n\mathbf{v}_n = \mathbf{0} \quad \Longrightarrow \quad \lambda_1 = \lambda_2 = \cdots = \lambda_n = 0.$$

That is, the *only* way to combine them back to the zero vector is the trivial way. If some non-trivial combination also reaches $\mathbf{0}$, the vectors are **dependent** — meaning at least one vector can be written as a linear combination of the others (it "adds no new information").
Practical test (as taught in class): stack the vectors as rows/columns of a matrix and row-reduce. If every column has a pivot (equivalently, if the determinant of a square matrix is non-zero), the vectors are independent.

Worked Example

$(1, 2)$ and $(2, 4)$ are dependent, since the second is exactly twice the first — $2 \cdot (1, 2) - 1 \cdot (2, 4) = \mathbf{0}$ using non-zero scalars $(2, -1)$.
Class also used an intuitive (non-strictly-vector) analogy: the R, G, B colour channels are "independent" building blocks from which every visible colour can be mixed; similarly, the 26 letters of the alphabet are independent enough to generate every English word.

Solved Question

Q. Are $\mathbf{v}_1 = (1, 2, 4),\ \mathbf{v}_2 = (5, 6, 8),\ \mathbf{v}_3 = (1, 1, 9)$ linearly independent? (This is the exact numeric example used in class to illustrate the rank test.)

Solution. Form the matrix with these as columns and compute its determinant:

$$M = \begin{pmatrix} 1 & 5 & 1 \\ 2 & 6 & 1 \\ 4 & 8 & 9 \end{pmatrix}$$

Expanding along row 1:

$$\det M = 1(6 \cdot 9 - 1 \cdot 8) - 5(2 \cdot 9 - 1 \cdot 4) + 1(2 \cdot 8 - 6 \cdot 4) = 1(46) - 5(14) + 1(-8) = 46 - 70 - 8 = -32.$$

Since $\det M = -32 \neq 0$, the three vectors are linearly independent.
Answer: Yes, independent (and since there are exactly 3 of them in $\mathbb{R}^3$, they also form a basis of $\mathbb{R}^3$ — see Section 1.4).

Real-World / ML Application

Checking whether the rows of a neural-network weight matrix are linearly independent tells you whether the layer is using its full representational capacity. This is precisely the mathematical basis of **LoRA** (Low-Rank Adaptation) for fine-tuning large language models, previewed in class: it assumes most directions in a weight update are *redundant* (dependent), so the update can be compressed into a small number of truly independent directions.

1.4 Basis and Dimension

Concept

A **basis** of a vector space is a set of vectors that is
(i) linearly independent, **and**
(ii) spans the entire space.
Both conditions are required — independence alone is not enough. The number of vectors in a basis is the **dimension** of the space.
The **standard basis** of $\mathbb{R}^2$ is $(1, 0), (0, 1)$; of $\mathbb{R}^3$ it is $(1, 0, 0), (0, 1, 0), (0, 0, 1)$.

Worked Example

Why both conditions matter (the exact caution raised in class): $(1, 0, 0)$ and $(0, 1, 0)$ in $\mathbb{R}^3$ are independent, but they only span the xy-plane (a 2-D slice of $\mathbb{R}^3$) — they can never reach a point with a non-zero z-coordinate. So they are *not* a basis for $\mathbb{R}^3$, only for that plane.
The three vectors from Section 1.3, $(1, 2, 4), (5, 6, 8), (1, 1, 9)$, *are* independent **and** there are exactly 3 of them in $\mathbb{R}^3$ (matching the dimension) — so they automatically span all of $\mathbb{R}^3$ and form a valid (non-standard) basis.

Solved Question

Q. Do $\mathbf{u}_1 = (1, 0, 0)$, $\mathbf{u}_2 = (0, 1, 0)$, $\mathbf{u}_3 = (1, 1, 0)$ form a basis of $\mathbb{R}^3$?
Solution. Check independence first: $\mathbf{u}_1 + \mathbf{u}_2 = (1, 1, 0) = \mathbf{u}_3$, so $\mathbf{u}_3$ is a linear combination of the other two — the set is *dependent*.
Answer: No. (Geometrically, all three vectors lie in the plane $z = 0$, so they can span only that 2-D plane, never the full $\mathbb{R}^3$.)

Real-World / ML Application

Principal Component Analysis (Sessions 12–13, previewed in class) is exactly the search for a *better basis* for your data — one where most of the "information" (variance) concentrates in the first few basis vectors, so that projecting onto only those few directions still preserves most of the structure (the professor's own

3-D → 2-D shape-flattening demo is exactly this idea).

1.5 Rank

Concept

The **rank** of a matrix is the number of linearly independent rows (equivalently, columns) it has — i.e., the dimension of the span of its rows/columns. It is found in practice by row-reducing to REF/RREF and counting pivot columns.

Worked Example

The Netflix analogy used in class: if a data matrix of 1 million user-preference vectors has rank 100, only 100 genuinely different ("independent") taste-directions exist among those million users; every other user's preferences are just a linear combination of those 100 archetypes.

Solved Question

Q. What is the rank of $M = \begin{pmatrix} 1 & 5 & 1 \\ 2 & 6 & 1 \\ 4 & 8 & 9 \end{pmatrix}$ (the matrix from Section 1.3)?

Solution. We already found $\det M = -32 \neq 0$. A non-zero determinant for a square matrix means it is invertible, which happens if and only if the matrix is *full rank*.
Answer: $\operatorname{rank}(M) = 3$ (full rank).

Real-World / ML Application

Real-world user–item interaction matrices (Netflix, Amazon, Spotify) are extremely large but have a *low effective rank* — a handful of latent "taste dimensions" explain almost all the variation. Matrix-factorization recommender systems (and the SVD you'll study in Session 5) exploit exactly this fact to compress and predict missing entries in these matrices.

PART II — Session 3: Analytic Geometry

Syllabus Link

Handout Session 3: *"Analytic Geometry – norms, inner products, lengths and distances, angles and orthogonality, orthonormal basis"* (Module M2, Textbook T1 Sec 3.1–3.5).

2.1 Norms

Concept

A **norm** attaches a non-negative "length" to every vector. The two norms used in this course:

Euclidean (L_2): $\|\mathbf{x}\|_2 = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$ Manhattan (L_1): $\|\mathbf{x}\|_1 = |x_1| + |x_2| + \cdots + |x_n|$

Euclidean norm is the straight-line ("as the crow flies") distance; Manhattan norm is the "taxicab" distance — travel along axes only, then add up the legs. In general, $\|\mathbf{x}\|_2 \le \|\mathbf{x}\|_1$ always (a direct route is never longer than a zig-zag one).

Worked Example

For $\mathbf{x} = (1, 2, 3)$:

$$\|\mathbf{x}\|_1 = |1| + |2| + |3| = 6, \qquad \|\mathbf{x}\|_2 = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14} \approx 3.742.$$

As expected, $3.742 < 6$.

Solved Question

Q. Compute both norms of $\mathbf{x} = (3, -4, 0, 12)$.
Solution.

$$\|\mathbf{x}\|_1 = 3 + 4 + 0 + 12 = 19, \qquad \|\mathbf{x}\|_2 = \sqrt{9 + 16 + 0 + 144} = \sqrt{169} = 13.$$

Answer: $\|\mathbf{x}\|_1 = 19$, $\|\mathbf{x}\|_2 = 13$.

Real-World / ML Application

L_1 vs. L_2 regularization (Lasso vs. Ridge regression, covered later at Lecture 10 in this course): an L_1 penalty on model weights encourages *sparsity* (many weights become exactly zero — automatic feature selection), while an L_2 penalty shrinks all weights smoothly without zeroing them out. Manhattan distance is also the natural distance metric for grid-based path planning, e.g., warehouse robots that can only move along aisles.

2.2 Dot Product and Cosine Similarity

Concept

For vectors $\mathbf{a}, \mathbf{b}$:

$$\mathbf{a} \cdot \mathbf{b} = \mathbf{a}^T \mathbf{b} = \sum_i a_i b_i = \|\mathbf{a}\| \, \|\mathbf{b}\| \cos\theta$$

The *sign* tells you about relative direction:

- $\mathbf{a} \cdot \mathbf{b} > 0$: vectors point in a broadly *similar* direction ($\theta < 90^\circ$).
- $\mathbf{a} \cdot \mathbf{b} = 0$: vectors are *orthogonal* ($\theta = 90^\circ$).
- $\mathbf{a} \cdot \mathbf{b} < 0$: vectors point in *opposite* directions ($\theta > 90^\circ$).

Dividing out the lengths gives **cosine similarity**, $\cos\theta = \dfrac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{a}\|\|\mathbf{b}\|} \in [-1, 1]$, the standard way to measure similarity between two vectors regardless of their magnitude.

Worked Example

$\mathbf{a} = (1, 4, 1, 6)$, $\mathbf{b} = (2, 1, 3, 9)$ (the exact example used in class):

$$\mathbf{a}\cdot\mathbf{b} = 1(2) + 4(1) + 1(3) + 6(9) = 2 + 4 + 3 + 54 = 63.$$

$\|\mathbf{a}\| = \sqrt{1 + 16 + 1 + 36} = \sqrt{54} \approx 7.348$, $\|\mathbf{b}\| = \sqrt{4 + 1 + 9 + 81} = \sqrt{95} \approx 9.747$.

$$\cos\theta = \frac{63}{7.348 \times 9.747} \approx \frac{63}{71.63} \approx 0.880 \quad (\theta \approx 28.4^\circ).$$

A high positive cosine similarity $\Rightarrow$ the two vectors point in a very similar direction.

Solved Question

Q. Two short documents are represented as term-frequency vectors $\mathbf{t}_1 = (2, 0, 1)$ and $\mathbf{t}_2 = (1, 1, 1)$. Compute their cosine similarity.
Solution.

$$\mathbf{t}_1\cdot\mathbf{t}_2 = 2(1) + 0(1) + 1(1) = 3, \qquad \|\mathbf{t}_1\| = \sqrt{4 + 0 + 1} = \sqrt{5} \approx 2.236, \qquad \|\mathbf{t}_2\| = \sqrt{1 + 1 + 1} = \sqrt{3} \approx 1.732.$$

$$\cos\theta = \frac{3}{2.236 \times 1.732} = \frac{3}{3.873} \approx 0.775.$$

Answer: $\cos\theta \approx 0.775$ — the two documents are fairly similar.

Real-World / ML Application

Search engines and recommender systems rank documents/items by the cosine similarity between TF-IDF or embedding vectors. The word-embedding visualization shown in class (words like "geography", "Karachi", "bordered" clustering together) is exactly a cosine-similarity-based neighbourhood in embedding space — the same principle behind semantic search in modern LLM-powered applications (e.g., retrieval-augmented generation).

2.3 Inner Product (Generalized Dot Product)

Concept

The ordinary dot product is $\mathbf{x}^T\mathbf{y} = \mathbf{x}^T I\mathbf{y}$ — a dot product taken with the identity matrix "hidden" in the middle. Replacing I with a symmetric, positive-definite matrix A gives the **inner product**

$$\langle \mathbf{x}, \mathbf{y}\rangle_A = \mathbf{x}^T A\mathbf{y},$$

which measures similarity *after* A has transformed the space. Every matrix represents some transformation (rotation, scaling, or both); the inner product asks: "how similar are $\mathbf{x}$ and $\mathbf{y}$ once A has reshaped the space they live in?"

Worked Example

Let $A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$, $\mathbf{x} = (1, 2)$, $\mathbf{y} = (1, -1)$.
Ordinary dot product: $\mathbf{x}\cdot\mathbf{y} = 1(1) + 2(-1) = -1$ (not orthogonal in the usual sense).
Inner product: first $A\mathbf{y} = (2(1), 1(-1)) = (2, -1)$, then $\mathbf{x}\cdot(A\mathbf{y}) = 1(2) + 2(-1) = 0$.

So under the A-weighted geometry, $\mathbf{x}$ and $\mathbf{y}$ *are* orthogonal, even though they weren't in the ordinary sense — "closeness" is always relative to which space you're measuring it in.

Solved Question

Q. With $A = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$, $\mathbf{x} = (1, 1)$, $\mathbf{y} = (1, -1)$, compute both the ordinary dot product and $\langle \mathbf{x}, \mathbf{y} \rangle_A$.
Solution. Ordinary: $\mathbf{x} \cdot \mathbf{y} = 1(1) + 1(-1) = 0$ (orthogonal in the usual Euclidean sense).
Weighted: $A\mathbf{y} = (3(1), 1(-1)) = (3, -1)$, so $\langle \mathbf{x}, \mathbf{y} \rangle_A = 1(3) + 1(-1) = 2$.
Answer: Ordinary dot product $= 0$, but $\langle \mathbf{x}, \mathbf{y} \rangle_A = 2 \neq 0$ — the opposite situation from the worked example above: two vectors that *are* orthogonal in the standard sense are *not* orthogonal once the space is reshaped by A.

Real-World / ML Application

The **Mahalanobis distance**, used heavily in anomaly/fraud detection, is exactly this generalized inner product with $A = \Sigma^{-1}$ (the inverse covariance matrix) — it measures distance while correctly accounting for correlations and different scales among features. **Kernel methods** in Support Vector Machines (Lecture 15–16 in this course) replace the ordinary dot product with a generalized inner product (a kernel) to measure similarity in an implicitly transformed feature space.

2.4 Positive Definite and Positive Semi-Definite Matrices

Concept

A symmetric matrix A is **positive definite** if $\mathbf{x}^T A\mathbf{x} > 0$ for every non-zero vector $\mathbf{x}$ (equivalently, all its eigenvalues are positive); it is **positive semi-definite** if $\mathbf{x}^T A\mathbf{x} \geq 0$. **Practical test taught before eigenvalues are available:** plug in a general vector $\mathbf{x} = (x_1, \ldots, x_n)$, expand $\mathbf{x}^T A\mathbf{x}$, and check the resulting expression can never go negative.

Worked Example

$A = \begin{pmatrix} 4 & 0 \\ 0 & 3 \end{pmatrix}$. For $\mathbf{x} = (x_1, x_2)$:

$$\mathbf{x}^T A\mathbf{x} = 4x_1^2 + 3x_2^2 \geq 0,$$

equal to zero only when $x_1 = x_2 = 0$. Hence A is positive definite.

Solved Question

Q. Is $A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ positive definite?
Solution.

$$\mathbf{x}^T A\mathbf{x} = 2x_1^2 - x_1x_2 - x_2x_1 + 2x_2^2 = 2x_1^2 + 2x_2^2 - 2x_1x_2.$$

Complete the square: $(x_1 - x_2)^2 = x_1^2 - 2x_1x_2 + x_2^2$, so

$$\mathbf{x}^T A\mathbf{x} = x_1^2 + x_2^2 + (x_1 - x_2)^2,$$

a sum of squares — always ≥ 0, and $= 0$ only if $x_1 = x_2 = 0$.
Answer: Yes, positive definite. (Cross-check with eigenvalues, once you know how to find them from Section 3.3: $\det(A - \lambda I) = (2 - \lambda)^2 - 1 = 0 \Rightarrow \lambda = 1, 3$ — both positive, confirming the result.)

Real-World / ML Application

If the Hessian of a loss function is positive definite at a point, the loss surface is locally convex ("bowl-shaped") there — gradient descent is then guaranteed to roll straight down to the minimum, exactly the "marble in a bowl" analogy used in class, instead of getting stuck at a saddle point. Covariance matrices are always positive semi-definite by construction, which is why PCA (built on the covariance matrix, Sessions 12–13) always produces real, non-negative eigenvalues.

2.5 The Cauchy–Schwarz Inequality

Concept

For any vectors $\mathbf{x}, \mathbf{y}$:

$$|\langle \mathbf{x}, \mathbf{y} \rangle| \le \|\mathbf{x}\| \, \|\mathbf{y}\|.$$

This follows immediately from $\cos\theta \in [-1, 1]$ in $\mathbf{x} \cdot \mathbf{y} = \|\mathbf{x}\|\|\mathbf{y}\| \cos\theta$. Equality holds exactly when $\mathbf{x}$ and $\mathbf{y}$ are parallel.

Worked Example

Reusing $\mathbf{a} = (1, 4, 1, 6),\ \mathbf{b} = (2, 1, 3, 9)$ from Section 2.2: $|\mathbf{a} \cdot \mathbf{b}| = 63$, and $\|\mathbf{a}\|\|\mathbf{b}\| \approx 71.63$. Indeed $63 \le 71.63$. ✓

Solved Question

Q. Verify Cauchy–Schwarz for $\mathbf{a} = (3, 4),\ \mathbf{b} = (4, 3)$.
Solution. $\mathbf{a} \cdot \mathbf{b} = 3(4) + 4(3) = 24$. $\|\mathbf{a}\| = \|\mathbf{b}\| = 5$, so $\|\mathbf{a}\|\|\mathbf{b}\| = 25$.
Answer: $24 \le 25$ ✓. (The two vectors are close to — but not exactly — parallel, so the inequality is close to, but not, an equality; $\cos\theta = 24/25 = 0.96$.)

Real-World / ML Application

Cauchy–Schwarz is used to derive the margin bound in Support Vector Machines (Lecture 15 in this course) and to prove that a Pearson correlation coefficient always lies in $[-1, 1]$ — a fact used every time you interpret a correlation matrix during exploratory data analysis.

2.6 Orthogonality, Orthonormal Vectors, and Orthonormal Matrices

Concept

Two vectors are **orthogonal** ($\mathbf{x} \perp \mathbf{y}$) iff $\mathbf{x} \cdot \mathbf{y} = 0$. **Orthonormal** vectors are orthogonal *and* unit length. A matrix Q is **orthogonal (orthonormal)** if its columns are orthonormal, equivalently $Q^T Q = I$, so $Q^{-1} = Q^T$. Orthogonal matrices represent pure **rotations** (or reflections) of space — crucially, they **preserve lengths and angles**.

Worked Example

$(4, 0)$ and $(0, 3)$: dot product $= 4(0) + 0(3) = 0$ — orthogonal. Normalizing (dividing each by its own length, 4 and 3) gives $(1, 0)$ and $(0, 1)$: orthonormal, and in fact the standard basis.

Rotation example (30°), as done in class: $R = \begin{pmatrix} \cos 30^\circ & -\sin 30^\circ \\ \sin 30^\circ & \cos 30^\circ \end{pmatrix} = \begin{pmatrix} 0.866 & -0.5 \\ 0.5 & 0.866 \end{pmatrix}$. Applying to $(1, 2)$:

$$R(1, 2) = (0.866(1) - 0.5(2),\ 0.5(1) + 0.866(2)) = (-0.134,\ 2.232).$$

Length before: $\sqrt{1^2 + 2^2} = \sqrt{5} \approx 2.236$. Length after: $\sqrt{(-0.134)^2 + (2.232)^2} \approx 2.236$ — unchanged, exactly as an orthogonal (rotation) matrix should behave.

Solved Question

Q. Show that $Q = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ is an orthogonal matrix.
Solution. Columns: $\mathbf{q}_1 = \frac{1}{\sqrt{2}}(1,1)$, $\mathbf{q}_2 = \frac{1}{\sqrt{2}}(1,-1)$.

$$\mathbf{q}_1 \cdot \mathbf{q}_1 = \tfrac{1}{2} + \tfrac{1}{2} = 1, \qquad \mathbf{q}_2 \cdot \mathbf{q}_2 = \tfrac{1}{2} + \tfrac{1}{2} = 1, \qquad \mathbf{q}_1 \cdot \mathbf{q}_2 = \tfrac{1}{2} - \tfrac{1}{2} = 0.$$

Both columns are unit length and mutually orthogonal.
Answer: Yes, Q is orthogonal (in fact, this specific Q is a 45° reflection/rotation matrix that appears again in the SVD practice problems in Appendix A).

Real-World / ML Application

Rotation matrices are the backbone of computer graphics and robotics (camera views, robot-arm joints). More importantly for this course: in both the Spectral Decomposition and SVD (Session 5 preview), the eigenvector/singular-vector matrices are exactly orthogonal matrices — they "rotate" the data into a new coordinate system *without distorting distances*, which is precisely why PCA preserves the geometric structure of data while re-expressing it in a more informative basis.

2.7 Gram–Schmidt Orthogonalization (Preview)

Concept

Given linearly independent vectors, the **Gram–Schmidt process** converts them into an orthogonal (and, after normalizing, orthonormal) set spanning the *same* space. Basic two-vector idea (as previewed in class, to be developed fully in the next session): project $\mathbf{v}_2$ onto $\mathbf{v}_1$, then subtract that projection from $\mathbf{v}_2$ — the remainder is automatically perpendicular to $\mathbf{v}_1$:

$$\mathbf{u}_1 = \mathbf{v}_1, \qquad \mathbf{u}_2 = \mathbf{v}_2 - \frac{\mathbf{v}_2 \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1}\,\mathbf{u}_1.$$

Worked Example

$\mathbf{v}_1 = (1,1)$, $\mathbf{v}_2 = (2,0)$. Projection factor: $\dfrac{\mathbf{v}_2 \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} = \dfrac{2(1)+0(1)}{1+1} = \dfrac{2}{2} = 1$, so the projection is $1 \cdot (1,1) = (1,1)$.

$$\mathbf{u}_2 = (2,0) - (1,1) = (1,-1).$$

Check: $\mathbf{u}_1 \cdot \mathbf{u}_2 = 1(1) + 1(-1) = 0$ ✓— perpendicular, as guaranteed.

Solved Question

Q. Orthogonalize $\mathbf{v}_1 = (1,0,1)$, $\mathbf{v}_2 = (1,1,0)$ (first Gram–Schmidt step).
Solution. $\mathbf{u}_1 \cdot \mathbf{u}_1 = 1+0+1 = 2$, $\mathbf{v}_2 \cdot \mathbf{u}_1 = 1(1)+1(0)+0(1) = 1$. Projection factor $= 1/2$, so projection $= \frac{1}{2}(1,0,1) = (0.5,0,0.5)$.

$$\mathbf{u}_2 = (1,1,0) - (0.5,0,0.5) = (0.5,\ 1,\ -0.5).$$

Check: $\mathbf{u}_1 \cdot \mathbf{u}_2 = 1(0.5) + 0(1) + 1(-0.5) = 0$ ✓.
Answer: $\mathbf{u}_1 = (1,0,1)$, $\mathbf{u}_2 = (0.5,1,-0.5)$, an orthogonal pair spanning the same plane as $\mathbf{v}_1, \mathbf{v}_2$.

Real-World / ML Application

Gram–Schmidt is the basis of **QR decomposition**, used to solve least-squares regression numerically stably (much better conditioned than solving the normal equations directly). This is exactly the subject of Lab 1 and Lab 2 in the course handout: solving $A\mathbf{x} = \mathbf{b}$ and studying how solution accuracy depends on the *condition number* of A, and the Gram–Schmidt orthogonalization lab.

PART III — Session 4: Matrix Decomposition – I

Syllabus Link

Handout Session 4: *"Matrix Decomposition – I: Determinant and Trace, Eigenvalues and Eigenvectors, Cholesky Decomposition"* (Module M3, Textbook T1 Sec 4.1–4.3). *Cholesky decomposition was not yet reached live in these recordings — see the syllabus map in Part V.*

Prof. Saurabh framed this entire block with a single analogy: just as the number 12 can be *factorized* into simple building blocks $2 \times 2 \times 3$, a matrix can be factorized into simple geometric transformations (rotation, scaling). Determinants and eigenvalues are the tools that make this factorization possible.

3.1 Determinants

Concept

The **determinant** of a matrix measures the (signed) factor by which the matrix scales area (in 2-D) or volume (in 3-D and higher). $\det(A) = 0$ means the transformation *collapses* space onto a lower dimension, so no inverse exists.
Cofactor expansion (recursive formula): along any row i (or column j),

$$\det(A) = \sum_j a_{ij} C_{ij}, \qquad C_{ij} = (-1)^{i+j} M_{ij},$$

where M_{ij} (the **minor**) is the determinant of the submatrix left after deleting row i and column j. The familiar 2×2 case, $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$, is this same formula applied once. **Tip taught in class:** expand along whichever row/column has the most zeros — it eliminates entire terms instantly.

Worked Example

$$A = \begin{pmatrix} 2 & 0 & 1 \\ 3 & 1 & 4 \\ 1 & 0 & 5 \end{pmatrix}$$

Column 2 has two zeros, so expand there ($a_{12} = 0,\ a_{22} = 1,\ a_{32} = 0$):

$$\det(A) = a_{22} \cdot (-1)^{2+2} M_{22} = 1 \cdot \det \begin{pmatrix} 2 & 1 \\ 1 & 5 \end{pmatrix} = 1(10 - 1) = 9.$$

Cross-check by expanding along row 1 instead:

$$\det(A) = 2 \det \begin{pmatrix} 1 & 4 \\ 0 & 5 \end{pmatrix} - 0 + 1 \det \begin{pmatrix} 3 & 1 \\ 1 & 0 \end{pmatrix} = 2(5) + 1(-1) = 10 - 1 = 9. \checkmark$$

Solved Question

Q. Compute $\det(B)$ for $B = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{pmatrix}$.

Solution. Expand along column 1 ($a_{11} = 1,\ a_{21} = 0,\ a_{31} = 1$):

$$\det(B) = 1 \cdot (+1) \det \begin{pmatrix} 4 & 5 \\ 0 & 6 \end{pmatrix} + 0 + 1 \cdot (+1) \det \begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix} = 1(24 - 0) + 1(10 - 12) = 24 - 2 = 22.$$

Cross-check along row 1: $1 \det \begin{pmatrix} 4 & 5 \\ 0 & 6 \end{pmatrix} - 2 \det \begin{pmatrix} 0 & 5 \\ 1 & 6 \end{pmatrix} + 3 \det \begin{pmatrix} 0 & 4 \\ 1 & 0 \end{pmatrix} = 24 - 2(-5) + 3(-4) = 24 + 10 - 12 =$

22. ✓
Answer: $\det(B) = 22$.

Real-World / ML Application

In linear regression, $\det(X^T X) = 0$ signals **multicollinearity** (redundant/dependent features) — the normal equations become unsolvable. Ridge regression fixes this by adding λI, i.e., solving $(X^T X + \lambda I)^{-1}$, deliberately making the determinant non-zero again. The determinant of a Jacobian also appears in the change-of-variables formula used inside **normalizing flows**, a generative-modelling technique.

3.2 Trace

Concept

The **trace** of A is the sum of its diagonal entries, $\mathrm{tr}(A) = \sum_i a_{ii}$. Once eigenvalues are introduced (Section 3.3), a key fact emerges: $\mathrm{tr}(A) = \sum_i \lambda_i$ — the trace equals the sum of the eigenvalues. This gives a fast sanity check on eigenvalue calculations without solving the full characteristic equation.

Worked Example

For $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ (worked fully in Section 3.3, eigenvalues $5, 2$): $\mathrm{tr}(A) = 4 + 3 = 7 = 5 + 2$. ✓

Real-World / ML Application

The trace of a covariance matrix equals the *total variance* in a dataset. In PCA (Sessions 12–13), the "% variance explained" by keeping the top k principal components is computed as (sum of top k eigenvalues) ÷ (trace of the covariance matrix) — this is exactly how you decide how many components to keep.

3.3 Eigenvalues and Eigenvectors

Concept

For a square matrix A, a non-zero vector $\mathbf{v}$ is an **eigenvector** with **eigenvalue** λ if

$$A\mathbf{v} = \lambda\mathbf{v}$$

— i.e., A only stretches/shrinks $\mathbf{v}$, never rotates it off its own line. Geometrically (as animated in class): most vectors get pushed off their original direction by a transformation, but eigenvectors are the special directions that survive, merely scaled.
To find them: solve the **characteristic equation** $\det(A - \lambda I) = 0$ for λ, then for each λ solve $(A - \lambda I)\mathbf{v} = \mathbf{0}$ for $\mathbf{v}$.

Worked Example

$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ (worked in class).

$$\det(A - \lambda I) = (4 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 7\lambda + 10 = (\lambda - 5)(\lambda - 2) = 0 \;\Rightarrow\; \lambda_1 = 5,\; \lambda_2 = 2.$$

Eigenvector for $\lambda = 5$: $(A - 5I)\mathbf{v} = \begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix}\mathbf{v} = \mathbf{0} \Rightarrow -v_1 + v_2 = 0 \Rightarrow \mathbf{v} = (1, 1)$.

Eigenvector for $\lambda = 2$: $(A - 2I)\mathbf{v} = \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix}\mathbf{v} = \mathbf{0} \Rightarrow 2v_1 + v_2 = 0 \Rightarrow \mathbf{v} = (1, -2)$.

Verify: $A(1, -2) = (4-2,\ 2-6) = (2, -4) = 2(1, -2)$. ✓ Also $\mathrm{tr}(A) = 7 = 5+2$ and $\det(A) = 10 = 5 \times 2$.

✓

Solved Question

Q. Find the eigenvalues and eigenvectors of $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ (a *symmetric* matrix — this exact matrix reappears in the Spectral Decomposition preview).
Solution. $\det(A - \lambda I) = (2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda-1)(\lambda-3) = 0 \Rightarrow \lambda = 1, 3$.
$\lambda = 3$: $(A - 3I)\mathbf{v} = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}\mathbf{v} = 0 \Rightarrow v_1 = v_2 \Rightarrow \mathbf{v} = (1, 1)$.
$\lambda = 1$: $(A - I)\mathbf{v} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\mathbf{v} = 0 \Rightarrow v_2 = -v_1 \Rightarrow \mathbf{v} = (1, -1)$.
Answer: $\lambda = 3,\ \mathbf{v} = (1, 1)$; $\lambda = 1,\ \mathbf{v} = (1, -1)$. **Notice:** $(1, 1) \cdot (1, -1) = 0$ — the two eigenvectors are *orthogonal*. This is not a coincidence: it always happens for symmetric matrices, and is exactly the Spectral Theorem previewed for Session 5.

Worked Example

A note on complex eigenvalues (mentioned in class): an orthogonal (pure-rotation) matrix has *no* real eigenvector, since a rotation never leaves any real direction unchanged. Its eigenvalues/eigenvectors work out to be **complex** numbers (of the form $a \pm bi$). Practice Problem 3 in Appendix A works through exactly such a case in full.

Real-World / ML Application

Eigenvectors identify "the directions along which a transformation acts simplest." As highlighted in class: Google's original **PageRank** algorithm is the dominant eigenvector of the web's link matrix; computer-vision corner detectors (Harris corner detector) use the eigenvalues of a local image structure tensor to flag sharp intensity changes (edges); civil/mechanical engineers find principal stress directions in a material via eigen-analysis of the stress tensor. Most importantly for this course: **PCA** (Sessions 12–13) is nothing but the eigen-decomposition of a covariance matrix.

3.4 The Power Method (Finding Eigenvalues Numerically)

Concept

For large matrices, solving $\det(A - \lambda I) = 0$ exactly (a degree-n polynomial) is computationally infeasible. The **Power Method** instead finds the *dominant* eigenvalue/eigenvector (the one with the largest magnitude) by repeated multiplication:

1. Start with any non-zero vector $\mathbf{x}_0$.
2. Iterate $\mathbf{x}_{k+1} = A\mathbf{x}_k$, then **normalize** by dividing by the largest-magnitude entry.
3. Repeat until the vector stops changing (converges).

The converged vector is the dominant eigenvector; the normalizing factor at convergence is the dominant eigenvalue. **Limitation noted in class:** the plain power method only ever finds the *largest* eigenvalue/eigenvector, not the others.

Worked Example

Apply the power method to $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ (whose true dominant eigenpair we already know: $\lambda = 5,\ \mathbf{v} = (1, 1)$), starting from $\mathbf{x}_0 = (1, 0)$:

Step	$A\mathbf{x}_k$	**Normalized (÷ largest entry)**
1	$(4,\ 2)$	$(1,\ 0.500)$
2	$(4.5,\ 3.5)$	$(1,\ 0.778)$
3	$(4.778,\ 4.333)$	$(1,\ 0.907)$
4	$(4.907,\ 4.722)$	$(1,\ 0.962)$

The second component is climbing steadily toward 1 (the true eigenvector $(1,1)$), and the normalizing factors $4,\ 4.5,\ 4.778,\ 4.907, \dots$ are climbing toward the true dominant eigenvalue 5.

Solved Question

Q. Perform two power-method iterations on $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ starting from $\mathbf{x}_0 = (1,0)$, and state which eigenvalue/eigenvector it is converging toward.

Solution.

$$\mathbf{x}_1 = A\mathbf{x}_0 = (2,1) \xrightarrow{\div 2} (1,0.5), \qquad \mathbf{x}_2 = A(1,0.5) = (2.5,2) \xrightarrow{\div 2.5} (1,0.8).$$

The second component is rising, $0.5 \to 0.8$, heading toward 1.

Answer: Converging toward eigenvector $(1,1)$ with eigenvalue $\lambda = 3$ — the larger of the two eigenvalues $\{1,3\}$ found in Section 3.3, exactly as the theory predicts.

Real-World / ML Application

The power method (and refinements of it) is exactly how **PageRank** is computed at web scale — with billions of pages, direct determinant-based methods are impossible, so the dominant eigenvector (page importance) is found via power iteration. It is also the numerical engine behind the built-in `eig` function used in this course's lab exercises (Lab 3: matrix powers via EVD; Lab 4: eigenvalues via the power method) and is the standard first algorithm taught before moving to the more refined numerical linear algebra routines that power libraries such as NumPy/SciPy.

PART IV — Preview: Session 5 (Matrix Decomposition – II)

Syllabus Link

Handout Session 5: *"Matrix Decomposition – II: Eigen-decomposition and Diagonalization, Singular Value Decomposition, Matrix Approximation"* (Module M3, Textbook T1 Sec 4.4–4.6). *This section is a brief preview, exactly as given in class — full derivations and fully-solved problems for these topics are in Appendix A (Problems 8–17).*

Prof. Saurabh closed the session by previewing what comes next, using the same "factorization" analogy: just as $12 = 2 \times 2 \times 3$, a well-behaved matrix factorizes into a sequence of simple geometric moves — **rotate** $\rightarrow$ **scale** $\rightarrow$ **rotate** (illustrated in class with a live 3-D animation of a cube being reshaped).

Concept

Eigen-decomposition / Diagonalization ($A = PDP^{-1}$): writes A in terms of its eigenvectors (columns of P) and eigenvalues (diagonal of D).
Spectral Decomposition ($A = PDP^T$): the special case for *symmetric* matrices, where P turns out to be *orthogonal* (a pure rotation) — this is why the symmetric-matrix eigenvectors found in Section 3.3 came out orthogonal to each other.
Singular Value Decomposition (SVD) ($A = U\Sigma V^T$): generalizes eigen-decomposition to *any* matrix, even non-square, non-symmetric ones.
Low-Rank Approximation: keeping only the top few eigenvalues/singular values reconstructs an almost-as-good approximation of A using far less storage.

Real-World / ML Application

This is exactly the mathematics behind:

- **Image compression** — the professor's own demo compressed a 60×60 image using rank-k approximations (rank 10, 19, 30, …), reconstructing a visually near-identical image while discarding most of the data, precisely the way WhatsApp compresses images before sending them.
- **LoRA / QLoRA**, the parameter-efficient fine-tuning techniques for today's large language models: instead of updating every weight in a huge matrix, only a small, low-rank update is learned — directly built on the low-rank approximation idea above.
- **PCA** (Sessions 12–13): the eigen-decomposition of the covariance matrix, used to find the directions of maximum variance in high-dimensional data.

PART V — Concept ↔ Syllabus Map

The table below maps every concept in these notes back to the official **Session Plan** (Contact Hours) and **Content Structure modules** in your course handout, so you can cross-check against the handout while revising.

Concept	Session	Module	Textbook (T1)
Linear combinations, span	2	M2	Sec 2.4–2.5
Linear independence, basis, dimension	2	M2	Sec 2.5–2.6
Rank	2	M2	Sec 2.6
Norms (Euclidean, Manhattan)	3	M2	Sec 3.1
Dot product, cosine similarity	3	M2	Sec 3.2
Inner product, positive definiteness	3	M2	Sec 3.2–3.3
Cauchy–Schwarz inequality	3	M2	Sec 3.3 (class notes)
Orthogonality, orthonormal basis	3	M2	Sec 3.4–3.5
Gram–Schmidt orthogonalization	3 (preview) / Lab 2	M2	Sec 3.5
Determinant & trace	4	M3	Sec 4.1
Eigenvalues & eigenvectors	4	M3	Sec 4.2
Power method	4 / Lab 4	M3	Sec 4.2 (numerical methods)
Cholesky decomposition	4 *(not yet covered live)*	M3	Sec 4.3
Eigen-decomposition & diagonalization	5 *(preview)*	M3	Sec 4.4
Singular Value Decomposition	5 *(preview)* / Lab 5	M3	Sec 4.5
Matrix (low-rank) approximation	5 *(preview)*	M3	Sec 4.6

Related lab exercises from the handout that put these ideas into code:

Lab	Objective	Session Reference
1	Solving $A\mathbf{x} = \mathbf{b}$ in Octave; accuracy vs. condition number of A	1, 2
2	Gram–Schmidt method of orthogonalization	2, 3
3	Powers of a 3×3 matrix: brute force vs. EVD (built-in `eig`)	4, 5
4	Calculating eigenvalues using the Power Method	4, 5
5	Finding the SVD of a 2×3 matrix using the built-in `eig` function	5

Appendix A — Practice Problem Set: Lectures 4 & 5 (Matrix Decompositions)

Reproduced from the supplied practice set, "MFML / MFDS · Decompositions Practice Set" (17 fully-solved problems), covering Lecture 4 (Eigenvalues & Eigenvectors) and Lecture 5 (Singular Value Decomposition). Problems 1–7 build directly on Section 3.3 of these notes (eigenvalues/eigenvectors); Problems 8–17 extend into diagonalization, spectral decomposition, and SVD — the Session 5 material previewed in Part IV. Read the two Concept boxes below first, then attempt each problem before reading its solution; every final result is boxed in green.

Concept 1: Algebraic vs. Geometric Multiplicity

Let λ be an eigenvalue of an $n \times n$ matrix A.

- **Algebraic multiplicity** $am(\lambda)$ — how many times λ occurs as a root of the characteristic polynomial $\det(A - \lambda I)$. (E.g., a factor $(\lambda - 3)^2$ means $am(3) = 2$.)
- **Geometric multiplicity** $gm(\lambda)$ — the number of independent eigenvectors for λ, i.e., $gm(\lambda) = \dim \ker(A - \lambda I) = n - \operatorname{rank}(A - \lambda I)$.

Key facts: (i) $1 \le gm(\lambda) \le am(\lambda)$ for every eigenvalue. (ii) A is diagonalizable $\iff$ $gm(\lambda) = am(\lambda)$ for every eigenvalue (enough eigenvectors to fill a basis). (iii) If $gm < am$ for some eigenvalue, A is **defective** and cannot be diagonalized.

One-line illustration: $\begin{pmatrix} 5 & 1 \\ 0 & 5 \end{pmatrix}$ has $\lambda = 5$ with $am = 2$ but $gm = 2 - \operatorname{rank}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = 2 - 1 = 1$ — defective. By contrast, $\begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix} = 5I$ has $am = gm = 2$ — diagonalizable (already diagonal).

Concept 2: The Eigenvalue Toolkit

If $A\mathbf{v} = \lambda \mathbf{v}$ ($\mathbf{v} \neq \mathbf{0}$), the eigenvalues transform predictably — the eigenvector stays the same:

- **Powers:** $A^k \mathbf{v} = \lambda^k \mathbf{v} \Rightarrow \lambda^k$ is an eigenvalue of A^k.
- **Polynomials:** if $p(t) = c_0 + c_1 t + \cdots + c_m t^m$, then $p(A)\mathbf{v} = p(\lambda)\mathbf{v}$.
- **Inverse:** if A is invertible ($\lambda \neq 0$), $A^{-1}\mathbf{v} = \frac{1}{\lambda}\mathbf{v}$.
- **Transpose:** A^T has the same eigenvalues as A.

Two consequences let you avoid brute-force matrix multiplication (writing eigenvalues as $\lambda_1, \ldots, \lambda_n$):

$$\operatorname{tr}(A) = \sum_i \lambda_i, \qquad \det(A) = \prod_i \lambda_i \quad \Longrightarrow \quad \operatorname{tr}(A^k) = \sum_i \lambda_i^k, \qquad \det(A^k) = (\det A)^k = \prod_i \lambda_i^k.$$

So to handle a high power, first find the eigenvalues, then work entirely with the scalars λ_i.

Problem 1 — Algebraic & Geometric Multiplicity (Warm-up) *[3 marks]*

For each matrix, state the eigenvalue(s), their algebraic multiplicity am and geometric multiplicity gm, and decide whether the matrix is diagonalizable.

$$A = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}, \qquad C = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 3 \end{pmatrix}.$$

Solution. Matrix A**:** triangular, so the only eigenvalue is $\lambda = 3$ with $am = 2$. $gm = 2 - \operatorname{rank}(A - 3I) = 2 - \operatorname{rank}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = 2 - 1 = 1$. Since $gm = 1 < 2 = am$, A is **not** diagonalizable.

Matrix B**:** $B = 3I$: $\lambda = 3$, $am = 2$, and $A - 3I = 0$ so $gm = 2$. Here $gm = am$, so B **is** diagonalizable (already diagonal).

Matrix C: triangular, $\lambda = 3$, $am = 3$. $C - 3I = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ has rank 2, so $gm = 3 - 2 = 1 < 3$. **Not** diagonalizable (a single Jordan block).

Answer

A: $\lambda = 3$, $am = 2$, $gm = 1$ — not diagonalizable. B: $\lambda = 3$, $am = gm = 2$ — diagonalizable. C: $\lambda = 3$, $am = 3$, $gm = 1$ — not diagonalizable.

Problem 2 — Eigen-Analysis of a 3×3 Matrix *[6 marks]*

Consider $A = \begin{pmatrix} 4 & 1 & 1 \\ 1 & 4 & 1 \\ 1 & 1 & 4 \end{pmatrix}$. (a) Find the characteristic polynomial and all eigenvalues with algebraic multiplicities. (b) For every eigenvalue, find a basis of its eigenspace and state the geometric multiplicity. (c) Decide, with reason, whether A is diagonalizable.

Solution. (a) Note $A = 3I + J$ where J is the all-ones matrix. Expanding $\det(A - \lambda I)$ gives $\det(A - \lambda I) = -(\lambda - 6)(\lambda - 3)^2$. So $\lambda = 6$ ($am = 1$) and $\lambda = 3$ ($am = 2$).

(b) For $\lambda = 6$: $(A - 6I)\mathbf{v} = 0$ reduces to $x = y = z$. Basis $\{(1, 1, 1)^T\}$, $gm = 1$.

For $\lambda = 3$: $(A - 3I)\mathbf{v} = 0$ collapses to the single equation $x + y + z = 0$ (all three rows become identical). Two free variables give basis $\{(-1, 1, 0)^T,\ (-1, 0, 1)^T\}$, $gm = 2$.

(c) For every eigenvalue, $gm = am$ ($1 = 1$ and $2 = 2$), so A is diagonalizable. (In fact A is symmetric, so the Spectral Theorem already guarantees this.)

Answer

$\det(A - \lambda I) = -(\lambda - 6)(\lambda - 3)^2$. $\lambda = 6$: mult. 1, eigenvector $(1, 1, 1)^T$; $\lambda = 3$: mult. 2, eigenspace $\{x + y + z = 0\}$, basis $(-1, 1, 0)^T, (-1, 0, 1)^T$. A is diagonalizable.

Problem 3 — Complex Eigenvalues of a Real Matrix *[4 marks]*

Find all eigenvalues and eigenvectors of $A = \begin{pmatrix} 3 & -2 \\ 4 & -1 \end{pmatrix}$, and explain geometrically why no real eigenvector exists.

Solution. $\det(A - \lambda I) = (3 - \lambda)(-1 - \lambda) + 8 = \lambda^2 - 2\lambda + 5 = 0 \Rightarrow \lambda = \dfrac{2 \pm \sqrt{4 - 20}}{2} = 1 \pm 2i.$

For $\lambda = 1 + 2i$: row 1 of $(A - \lambda I)\mathbf{v} = 0$ gives $(2 - 2i)x - 2y = 0 \Rightarrow y = (1 - i)x$. Taking $x = 1$: $\mathbf{v}_1 = (1,\ 1 - i)^T$. Its conjugate $\mathbf{v}_2 = (1,\ 1 + i)^T$ belongs to $\lambda = 1 - 2i$.

Geometric reason: the discriminant is negative, so the eigenvalues form a complex-conjugate pair. Such a real 2×2 matrix acts as a rotation combined with scaling; a rotation maps no non-zero real vector onto a multiple of itself, hence no real eigenvector exists.

Answer

$\lambda = 1 \pm 2i$ with eigenvectors $\mathbf{v} = (1,\ 1 \mp i)^T$. No real eigenvector exists because the eigenvalues are non-real (the map rotates the plane).

Problem 4 — Parametric Eigen-Analysis *[4 marks]*

Let $a, b, c \in \mathbb{R}$ with $b \neq 0$ and $A = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$. (a) Find the eigenvalues and an eigenvector for each. (b) State exactly when A is diagonalizable.

Solution. (a) Triangular $\Rightarrow \lambda_1 = a,\ \lambda_2 = c$.

$\lambda_1 = a$: $(A - aI) = \begin{pmatrix} 0 & b \\ 0 & c-a \end{pmatrix}$. Since $b \neq 0$, $y = 0$, x free $\Rightarrow \mathbf{v}_1 = (1, 0)^T$.

$\lambda_2 = c$ (assume $c \neq a$): $(A - cI) = \begin{pmatrix} a-c & b \\ 0 & 0 \end{pmatrix} \Rightarrow (a-c)x + by = 0 \Rightarrow \mathbf{v}_2 = (-b,\ a-c)^T$.

(b) If $a \neq c$: two distinct eigenvalues $\Rightarrow$ diagonalizable. If $a = c$: $\lambda = a$ has $am = 2$, but $(A - aI) = \begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix}$ has rank 1 (since $b \neq 0$), so $gm = 1 < 2$: not diagonalizable.

Answer

$\lambda_1 = a$ with $\mathbf{v}_1 = (1,0)^T$; $\lambda_2 = c$ with $\mathbf{v}_2 = (-b,\ a-c)^T$. A is diagonalizable $\iff$ $a \neq c$ (when $a = c$ and $b \neq 0$ it is defective).

Problem 5 — Is the 3×3 Matrix Diagonalizable? ***[4 marks]***

Decide whether $A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$ is diagonalizable.

Solution. Upper-triangular $\Rightarrow \lambda = 2$ ($am = 2$), $\lambda = 3$ ($am = 1$). $A - 2I = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ has rank 2, so its null space has dimension $3 - 2 = 1$: eigenspace $\operatorname{span}\{(1,0,0)^T\}$, $gm = 1$.

Since $gm = 1 < 2 = am$ for $\lambda = 2$, A cannot have a basis of eigenvectors.

Answer

Not diagonalizable: for $\lambda = 2$, $gm = 1 < am = 2$ (only one independent eigenvector $(1,0,0)^T$). The block $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ is a defective Jordan block.

Problem 6 — Powers Without Brute Force: Determinant and Trace ***[4 marks]***

$A = \begin{pmatrix} 2 & -1 & 1 \\ 1 & 0 & 3 \\ 1 & -1 & 4 \end{pmatrix}$, with characteristic polynomial $(\lambda - 1)(\lambda - 2)(\lambda - 3)$. Without forming any matrix power explicitly, compute (a) $\det(A^8)$, (b) $\operatorname{tr}(A^6)$, (c) $\operatorname{tr}(A^6 - 7A^2)$.

Solution. Eigenvalues: $\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 3$ (sanity check: $\operatorname{tr} A = 6 = 1 + 2 + 3$, $\det A = 6 = 1 \cdot 2 \cdot 3$).

(a) $\det(A^8) = (\det A)^8 = 6^8 = 1\,679\,616$.

(b) $\operatorname{tr}(A^6) = \sum_i \lambda_i^6 = 1^6 + 2^6 + 3^6 = 1 + 64 + 729 = 794$.

(c) Trace is linear: $\operatorname{tr}(A^6 - 7A^2) = \sum_i(\lambda_i^6 - 7\lambda_i^2) = 794 - 7(1 + 4 + 9) = 794 - 98 = 696$.

Answer

$\det(A^8) = 6^8 = 1\,679\,616$; $\operatorname{tr}(A^6) = 794$; $\operatorname{tr}(A^6 - 7A^2) = 696$.

Problem 7 — Eigenvalues of a Matrix Polynomial $p(A)$ ***[4 marks]***

A is 3×3 with eigenvalues $1, 2, 4$; define $B = A^2 - 5A + 6I$. (a) Show that if λ is an eigenvalue of A, then $\lambda^2 - 5\lambda + 6$ is an eigenvalue of B (same eigenvector). (b) Find the eigenvalues of B, then $\det(B)$, $\operatorname{tr}(B)$; is B invertible? (c) For which eigenvalues of A would B fail to be invertible?

Solution. (a) If $A\mathbf{v} = \lambda\mathbf{v}$, then $A^2\mathbf{v} = \lambda^2\mathbf{v}$, so $B\mathbf{v} = (\lambda^2 - 5\lambda + 6)\mathbf{v}$.

(b) $p(\lambda) = \lambda^2 - 5\lambda + 6$: $p(1) = 2$, $p(2) = 0$, $p(4) = 2$. So B has eigenvalues $\{2, 0, 2\}$: $\det(B) = 0$, $\mathrm{tr}(B) = 4$. Since 0 is an eigenvalue, B is **not invertible** (even though A itself is).

(c) $p(\lambda) = (\lambda - 2)(\lambda - 3)$, so B is singular iff A has eigenvalue 2 or 3. Here $\lambda = 2$ occurs.

Answer

$p(\lambda) = \lambda^2 - 5\lambda + 6$; eigenvalues of B are $2, 0, 2$. $\det(B) = 0$, $\mathrm{tr}(B) = 4$, so B is not invertible. B is singular iff A has eigenvalue 2 or 3.

Problem 8 — Eigendecomposition $A = PDP^{-1}$ ***[4 marks]***

Find an eigendecomposition of the non-symmetric matrix $A = \begin{pmatrix} 5 & -1 \\ 2 & 2 \end{pmatrix}$, giving P, D, P^{-1} explicitly, and verify.

Solution. $\det(A - \lambda I) = (5 - \lambda)(2 - \lambda) + 2 = \lambda^2 - 7\lambda + 12 = (\lambda - 3)(\lambda - 4) = 0 \Rightarrow \lambda_1 = 3, \lambda_2 = 4$.

$\lambda_1 = 3$: $(A - 3I) = \begin{pmatrix} 2 & -1 \\ 2 & -1 \end{pmatrix} \Rightarrow 2x = y \Rightarrow \mathbf{p}_1 = (1, 2)^T$.

$\lambda_2 = 4$: $(A - 4I) = \begin{pmatrix} 1 & -1 \\ 2 & -2 \end{pmatrix} \Rightarrow x = y \Rightarrow \mathbf{p}_2 = (1, 1)^T$.

$$P = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 3 & 0 \\ 0 & 4 \end{pmatrix}, \quad \det P = -1, \quad P^{-1} = \begin{pmatrix} -1 & 1 \\ 2 & -1 \end{pmatrix}.$$

Check: $PD = \begin{pmatrix} 3 & 4 \\ 6 & 4 \end{pmatrix}$, and $PDP^{-1} = \begin{pmatrix} 3 & 4 \\ 6 & 4 \end{pmatrix} \begin{pmatrix} -1 & 1 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} 5 & -1 \\ 2 & 2 \end{pmatrix} = A$. ✓

Answer

$A = PDP^{-1}$ with $P = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix}$, $D = \begin{pmatrix} 3 & 0 \\ 0 & 4 \end{pmatrix}$, $P^{-1} = \begin{pmatrix} -1 & 1 \\ 2 & -1 \end{pmatrix}$.

Problem 9 — Spectral Decomposition $A = PDP^T$ **of a Symmetric** 3×3 **Matrix** ***[6 marks]***

$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$. Find eigenvalues, an orthonormal eigenbasis, and write $A = PDP^T$ with P orthogonal.

Solution. Expanding $\det(A - \lambda I)$ along row 1: $(2 - \lambda)\left[(2 - \lambda)^2 - 2\right] = 0 \Rightarrow \lambda = 2$ or $(2 - \lambda)^2 = 2$, giving

$$\lambda_1 = 2 + \sqrt{2}, \qquad \lambda_2 = 2, \qquad \lambda_3 = 2 - \sqrt{2}.$$

$\lambda = 2$: $(A - 2I)\mathbf{v} = 0 \Rightarrow y = 0$, $x + z = 0 \Rightarrow \mathbf{v}_2 = (1, 0, -1)^T$.

$\lambda = 2 + \sqrt{2}$: row 1 gives $-\sqrt{2}x + y = 0 \Rightarrow y = \sqrt{2}x$; row 3 gives $x = z$. $\mathbf{v}_1 = (1, \sqrt{2}, 1)^T$.

$\lambda = 2 - \sqrt{2}$: similarly $\mathbf{v}_3 = (1, -\sqrt{2}, 1)^T$. The three eigenvectors are mutually orthogonal (guaranteed by the Spectral Theorem).

Normalizing ($\|\mathbf{v}_2\| = \sqrt{2}$, $\|\mathbf{v}_1\| = \|\mathbf{v}_3\| = 2$):

$$P = \begin{pmatrix} \frac{1}{2} & \frac{1}{\sqrt{2}} & \frac{1}{2} \\ \frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \\ \frac{1}{2} & -\frac{1}{\sqrt{2}} & \frac{1}{2} \end{pmatrix}, \qquad D = \mathrm{diag}(2 + \sqrt{2},\ 2,\ 2 - \sqrt{2}), \qquad A = PDP^T \ (P^TP = I).$$

Answer

Eigenvalues $2+\sqrt{2},\ 2,\ 2-\sqrt{2}$ with orthonormal eigenvectors $\frac{1}{2}(1,\sqrt{2},1)^T,\ \frac{1}{\sqrt{2}}(1,0,-1)^T,\ \frac{1}{2}(1,-\sqrt{2},1)^T$; $A = PDP^T$, P orthogonal.

Problem 10 — Matrix Powers via Diagonalization ***[4 marks]***

Derive a closed form for A^n and evaluate A^4, where $A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$.

Solution. $\det(A-\lambda I) = (2-\lambda)^2 - 1 = (\lambda-1)(\lambda-3) = 0 \Rightarrow \lambda_1 = 1, \lambda_2 = 3$. Orthonormal eigenvectors $\frac{1}{\sqrt{2}}(1,1)^T,\ \frac{1}{\sqrt{2}}(1,-1)^T$ give

$$P = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad D = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}, \qquad A = PDP^T.$$

Since P is orthogonal, $A^n = PD^nP^T$ (only the diagonal is raised):

$$A^n = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 3^n \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1+3^n & 1-3^n \\ 1-3^n & 1+3^n \end{pmatrix}.$$

For $n = 4$: $3^4 = 81$, so $A^4 = \frac{1}{2}\begin{pmatrix} 82 & -80 \\ -80 & 82 \end{pmatrix} = \begin{pmatrix} 41 & -40 \\ -40 & 41 \end{pmatrix}$.

Answer

$A^n = \frac{1}{2}\begin{pmatrix} 1+3^n & 1-3^n \\ 1-3^n & 1+3^n \end{pmatrix}$, $A^4 = \begin{pmatrix} 41 & -40 \\ -40 & 41 \end{pmatrix}$.

Problem 11 — Full SVD of a 3×2 Matrix ***[6 marks]***

Find the full SVD $A = U\Sigma V^T$ of $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$, including the third column of U via orthogonal completion.

Solution. $A^TA = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, eigenvalues $\lambda = 3, 1 \Rightarrow \sigma_1 = \sqrt{3},\ \sigma_2 = 1$, with $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1,1)^T,\ \mathbf{v}_2 = \frac{1}{\sqrt{2}}(1,-1)^T$.

$\mathbf{u}_i = \frac{1}{\sigma_i}A\mathbf{v}_i$: $\mathbf{u}_1 = \frac{1}{\sqrt{6}}(2,1,1)^T,\ \mathbf{u}_2 = \frac{1}{\sqrt{2}}(0,1,-1)^T$.

Orthogonal completion $\mathbf{u}_3$: need $\mathbf{u}_3 = (p,q,r)^T \perp \mathbf{u}_1, \mathbf{u}_2$. From $\mathbf{u}_2$: $q = r$; from $\mathbf{u}_1$: $2p+q+r = 0 \Rightarrow p = -q$. So $\mathbf{u}_3 = \frac{1}{\sqrt{3}}(-1,1,1)^T$.

$$U = \begin{pmatrix} 2/\sqrt{6} & 0 & -1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \\ 1/\sqrt{6} & -1/\sqrt{2} & 1/\sqrt{3} \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad V^T = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Answer

$\sigma_1 = \sqrt{3},\ \sigma_2 = 1$; $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1,1)^T,\ \mathbf{v}_2 = \frac{1}{\sqrt{2}}(1,-1)^T$; $\mathbf{u}_1 = \frac{1}{\sqrt{6}}(2,1,1)^T,\ \mathbf{u}_2 = \frac{1}{\sqrt{2}}(0,1,-1)^T,\ \mathbf{u}_3 = \frac{1}{\sqrt{3}}(-1,1,1)^T$.

Problem 12 — Full SVD of a 2×3 Matrix ***[6 marks]***

Find the SVD of $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$. *(Hint: it's quicker to get U and the singular values from the 2×2 matrix AA^T, then recover V from $\mathbf{v}_i = \frac{1}{\sigma_i}A^T\mathbf{u}_i$; the missing right vector spans $\ker(A^TA)$.)*

Solution. $AA^T = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \Rightarrow \lambda = 3, 1,\ \sigma_1 = \sqrt{3},\ \sigma_2 = 1$. $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1,1)^T,\ \mathbf{u}_2 = \frac{1}{\sqrt{2}}(1,-1)^T$.

$\mathbf{v}_i = \frac{1}{\sigma_i}A^T\mathbf{u}_i$: $\mathbf{v}_1 = \frac{1}{\sqrt{6}}(1,2,1)^T,\ \mathbf{v}_2 = \frac{1}{\sqrt{2}}(1,0,-1)^T$.

Third right vector (from $\ker A$, since A is 2×3): solving $A\mathbf{v} = 0$ gives $x + y = 0,\ y + z = 0 \Rightarrow \mathbf{v}_3 = \frac{1}{\sqrt{3}}(1,-1,1)^T$.

$$U = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad V = \begin{pmatrix} 1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \\ 2/\sqrt{6} & 0 & -1/\sqrt{3} \\ 1/\sqrt{6} & -1/\sqrt{2} & 1/\sqrt{3} \end{pmatrix}.$$

The single zero column of Σ reflects that A has rank 2 but 3 columns.

Answer

$\sigma_1 = \sqrt{3},\ \sigma_2 = 1$; $U = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$; $\mathbf{v}_1 = \frac{1}{\sqrt{6}}(1,2,1)^T,\ \mathbf{v}_2 = \frac{1}{\sqrt{2}}(1,0,-1)^T,\ \mathbf{v}_3 = \frac{1}{\sqrt{3}}(1,-1,1)^T$.

Problem 13 — SVD With a Zero Singular Value; SVD vs. Diagonalizability ***[6 marks]***

(a) Find the SVD of $A = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$. (b) Show A is not diagonalizable, yet its SVD still exists — explain why.

Solution. (a) $A^TA = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$, eigenvalues $4, 0 \Rightarrow \sigma_1 = 2,\ \sigma_2 = 0$ (rank $A = 1$). $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1,1)^T$ (for 4), $\mathbf{v}_2 = \frac{1}{\sqrt{2}}(1,-1)^T$ (for 0).

$\mathbf{u}_1 = \frac{1}{\sigma_1} A\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1,-1)^T$; $\mathbf{u}_2 \perp \mathbf{u}_1 \Rightarrow \mathbf{u}_2 = \frac{1}{\sqrt{2}}(1,1)^T$.

$$U = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ -1 & 1\end{pmatrix}, \quad \Sigma = \begin{pmatrix}2 & 0\\ 0 & 0\end{pmatrix}, \quad V = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}.$$

(b) Eigenvalues of A itself: $\det(A-\lambda I) = \lambda^2 = 0 \Rightarrow \lambda = 0$ ($am = 2$), but $A\mathbf{v} = 0$ has only the single direction $(1,-1)^T$ ($gm = 1$). Since $1 < 2$, A has no eigenbasis — not diagonalizable. Yet the SVD is built from A^TA and AA^T, which are always orthogonally diagonalizable (symmetric) — this is exactly why an SVD exists for **every** matrix, diagonalizable or not.

Answer

$\sigma_1 = 2,\ \sigma_2 = 0$; $U = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ -1 & 1\end{pmatrix}$, $\Sigma = \begin{pmatrix}2 & 0\\ 0 & 0\end{pmatrix}$, $V = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}$. A is not diagonalizable ($\lambda = 0$: $am = 2,\ gm = 1$).

Problem 14 — Best Rank-1 Approximation *[4 marks]*

For $A = \begin{pmatrix}2 & 2\\ -1 & 1\end{pmatrix}$, find the singular values and the best rank-1 approximation $A_1 = \sigma_1\mathbf{u}_1\mathbf{v}_1^T$.

Solution. $A^TA = \begin{pmatrix}5 & 3\\ 3 & 5\end{pmatrix}$, eigenvalues $8, 2 \Rightarrow \sigma_1 = 2\sqrt{2},\ \sigma_2 = \sqrt{2}$, with $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1,1)^T$ (for $\lambda = 8$).

$\mathbf{u}_1 = \frac{1}{\sigma_1}A\mathbf{v}_1 = \frac{1}{2\sqrt{2}} \cdot \frac{1}{\sqrt{2}}(4,0)^T = (1,0)^T$.

$$A_1 = \sigma_1\mathbf{u}_1\mathbf{v}_1^T = 2\sqrt{2}\begin{pmatrix}1\\ 0\end{pmatrix} \cdot \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\end{pmatrix} = \begin{pmatrix}2 & 2\\ 0 & 0\end{pmatrix}.$$

By the Eckart–Young theorem, this is the closest rank-1 matrix to A; the discarded part corresponds to $\sigma_2 = \sqrt{2}$.

Answer

$\sigma_1 = 2\sqrt{2},\ \sigma_2 = \sqrt{2}$; $A_1 = \begin{pmatrix}2 & 2\\ 0 & 0\end{pmatrix}$ with $\mathbf{u}_1 = (1,0)^T,\ \mathbf{v}_1 = \frac{1}{\sqrt{2}}(1,1)^T$.

Problem 15 — SVD as a Sum of Rank-1 Matrices *[4 marks]*

A matrix has SVD data: $\sigma_1 = 3,\ \mathbf{u}_1 = \frac{1}{\sqrt{2}}(1,1)^T,\ \mathbf{v}_1 = \frac{1}{\sqrt{2}}(1,1)^T$; $\sigma_2 = 1,\ \mathbf{u}_2 = \frac{1}{\sqrt{2}}(1,-1)^T,\ \mathbf{v}_2 = \frac{1}{\sqrt{2}}(1,-1)^T$. (a) Reconstruct $A = \sigma_1\mathbf{u}_1\mathbf{v}_1^T + \sigma_2\mathbf{u}_2\mathbf{v}_2^T$. (b) Give the best rank-1 approximation and the fraction of "energy" it keeps.

Solution. (a) $\sigma_1\mathbf{u}_1\mathbf{v}_1^T = \frac{3}{2}\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}$, $\sigma_2\mathbf{u}_2\mathbf{v}_2^T = \frac{1}{2}\begin{pmatrix}1 & -1\\ -1 & 1\end{pmatrix}$. Adding:

$$A = \frac{1}{2}\begin{pmatrix}3 & 3\\ 3 & 3\end{pmatrix} + \frac{1}{2}\begin{pmatrix}1 & -1\\ -1 & 1\end{pmatrix} = \begin{pmatrix}2 & 1\\ 1 & 2\end{pmatrix}.$$

(b) Keep only σ_1: $A_1 = \frac{3}{2}\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}$. Energy kept $= \dfrac{\sigma_1^2}{\sigma_1^2+\sigma_2^2} = \dfrac{9}{10} = 90\%$.

Answer

$A = \begin{pmatrix}2 & 1\\ 1 & 2\end{pmatrix}$; best rank-1 approximation $A_1 = \frac{3}{2}\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}$, keeping 90% of the energy.

Problem 16 — Spectral Norm of a Matrix ***[4 marks]***

Compute the spectral norm $\|A\|_2$ of $A = \begin{pmatrix} 4 & 3 \\ 0 & 5 \end{pmatrix}$, and state the corresponding right singular vector.

Solution. $\|A\|_2 = \sigma_{\max} = \sqrt{\lambda_{\max}(A^T A)}$. $A^T A = \begin{pmatrix} 16 & 12 \\ 12 & 34 \end{pmatrix}$: tr $= 50$, det $= 16(34) - 12^2 = 400$.

$$\lambda = \frac{50 \pm \sqrt{2500 - 1600}}{2} = \frac{50 \pm 30}{2} = 40 \text{ or } 10.$$

So $\sigma_{\max} = \sqrt{40} = 2\sqrt{10}$, $\sigma_{\min} = \sqrt{10}$. Right singular vector: $(A^T A - 40I)\mathbf{v} = \begin{pmatrix} -24 & 12 \\ 12 & -6 \end{pmatrix} \mathbf{v} = 0 \Rightarrow 2x = y \Rightarrow \mathbf{v}_1 = \frac{1}{\sqrt{5}}(1, 2)^T$.

Answer

$\|A\|_2 = \sigma_{\max} = 2\sqrt{10} \approx 6.32$, attained along $\mathbf{v}_1 = \frac{1}{\sqrt{5}}(1, 2)^T$ (the other singular value is $\sqrt{10}$).

Problem 17 — Capstone: EVD vs. SVD of One Matrix ***[6 marks]***

For $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$: (a) find its eigendecomposition (note the negative eigenvalue); (b) find its SVD; (c) explain precisely how the two are related.

Solution. (a) $\det(A - \lambda I) = (1 - \lambda)^2 - 4 = (\lambda - 3)(\lambda + 1) = 0 \Rightarrow \lambda_1 = 3, \lambda_2 = -1$. Orthonormal eigenvectors $\mathbf{q}_1 = \frac{1}{\sqrt{2}}(1, 1)^T$ (for 3), $\mathbf{q}_2 = \frac{1}{\sqrt{2}}(1, -1)^T$ (for -1). With $Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$: $A = Q \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix} Q^T$.

(b) $A^T A = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$, eigenvalues $9, 1 \Rightarrow \sigma_1 = 3, \sigma_2 = 1$ — note $\sigma_i = |\lambda_i|$. Right singular vectors are the same eigenvectors $\mathbf{v}_1 = \mathbf{q}_1$, $\mathbf{v}_2 = \mathbf{q}_2$, and

$$\mathbf{u}_1 = \tfrac{1}{3} A\mathbf{v}_1 = \tfrac{1}{\sqrt{2}}(1, 1)^T = \mathbf{v}_1, \qquad \mathbf{u}_2 = \tfrac{1}{1} A\mathbf{v}_2 = \tfrac{1}{\sqrt{2}}(-1, 1)^T = -\mathbf{v}_2.$$

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}, \qquad V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

(c) For a symmetric matrix, the singular values are the absolute values of the eigenvalues, $\sigma_i = |\lambda_i|$, and the right singular vectors are the eigenvectors. Where an eigenvalue is positive, $\mathbf{u}_i = \mathbf{v}_i$; where negative (here $\lambda_2 = -1$), the sign is absorbed into the left vector, $\mathbf{u}_i = -\mathbf{v}_i$. EVD and SVD coincide exactly when every eigenvalue is non-negative (a positive-semidefinite matrix); otherwise they differ only by these sign flips between U and V.

Answer

EVD: $A = Q \operatorname{diag}(3, -1) Q^T$, $Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. SVD: $\Sigma = \operatorname{diag}(3, 1) = \operatorname{diag}(|\lambda_i|)$, $V = Q$, $U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ (the negative eigenvalue flips the sign of $\mathbf{u}_2$ relative to $\mathbf{v}_2$).

End of Practice Problem Set — 17 problems, all topics from Lecture 4 & Lecture 5 (matrix decompositions).

Quick-Reference Formula Sheet

Topic	Key Formula
Linear combination	$\lambda_1\mathbf{v}_1 + \cdots + \lambda_n\mathbf{v}_n$
Independence test	$\lambda_1\mathbf{v}_1 + \cdots + \lambda_n\mathbf{v}_n = \mathbf{0} \Rightarrow$ all $\lambda_i = 0$
Basis	independent + spans the whole space
Euclidean norm	$\|\mathbf{x}\|_2 = \sqrt{\sum x_i^2}$
Manhattan norm	$\|\mathbf{x}\|_1 = \sum \|x_i\|$
Dot product	$\mathbf{a}\cdot\mathbf{b} = \sum a_i b_i = \|\mathbf{a}\|\|\mathbf{b}\|\cos\theta$
Cosine similarity	$\cos\theta = \mathbf{a}\cdot\mathbf{b}/(\|\mathbf{a}\|\|\mathbf{b}\|)$
Inner product	$\langle\mathbf{x},\mathbf{y}\rangle_A = \mathbf{x}^T A\mathbf{y}$
Positive definite	$\mathbf{x}^T A\mathbf{x} > 0\ \forall\mathbf{x}\neq 0 \iff$ all $\lambda_i > 0$
Cauchy–Schwarz	$\|\langle\mathbf{x},\mathbf{y}\rangle\| \leq \|\mathbf{x}\|\|\mathbf{y}\|$
Orthogonal matrix	$Q^TQ = I \Rightarrow Q^{-1} = Q^T$
Gram–Schmidt	$\mathbf{u}_2 = \mathbf{v}_2 - \frac{\mathbf{v}_2\cdot\mathbf{u}_1}{\mathbf{u}_1\cdot\mathbf{u}_1}\mathbf{u}_1$
Determinant (2×2)	$\det = ad - bc$
Cofactor expansion	$\det(A) = \sum_j a_{ij}(-1)^{i+j}M_{ij}$
Trace	$\mathrm{tr}(A) = \sum_i a_{ii} = \sum_i \lambda_i$
Eigen-equation	$A\mathbf{v} = \lambda\mathbf{v} \iff \det(A - \lambda I) = 0$
Diagonalizable	$gm(\lambda) = am(\lambda)$ for every λ
Eigendecomposition	$A = PDP^{-1}$
Spectral decomposition	$A = PDP^T$ (A symmetric, P orthogonal)
SVD	$A = U\Sigma V^T$ (U, V orthogonal, Σ diagonal ≥ 0)
Best rank-k approx.	$A_k = \sum_{i=1}^k \sigma_i\mathbf{u}_i\mathbf{v}_i^T$
Spectral norm	$\|A\|_2 = \sigma_{\max}$